

Module 2 Assignment

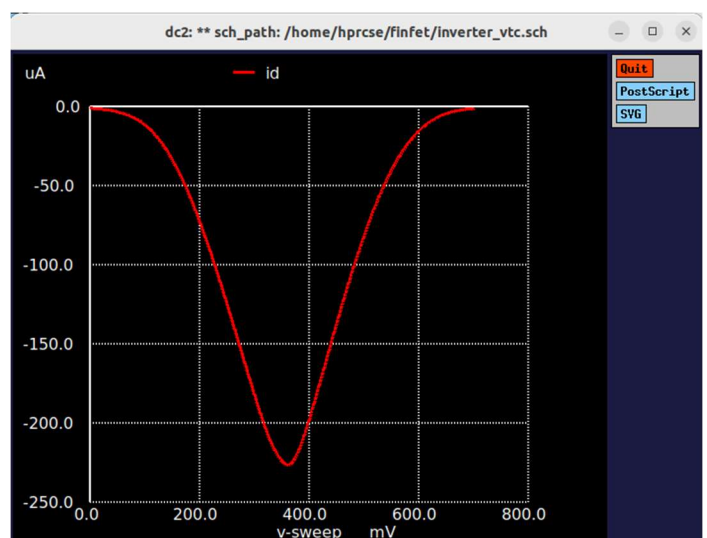
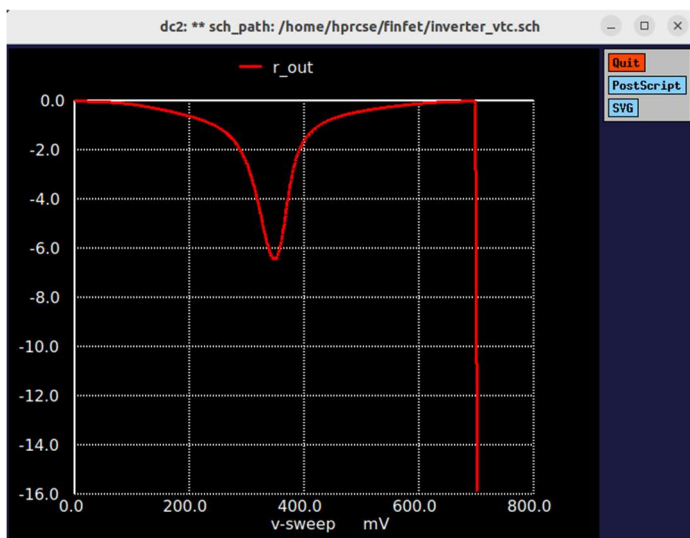
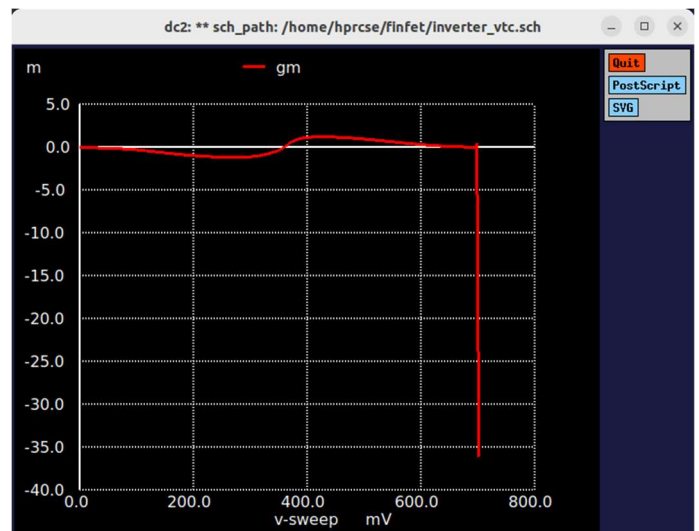
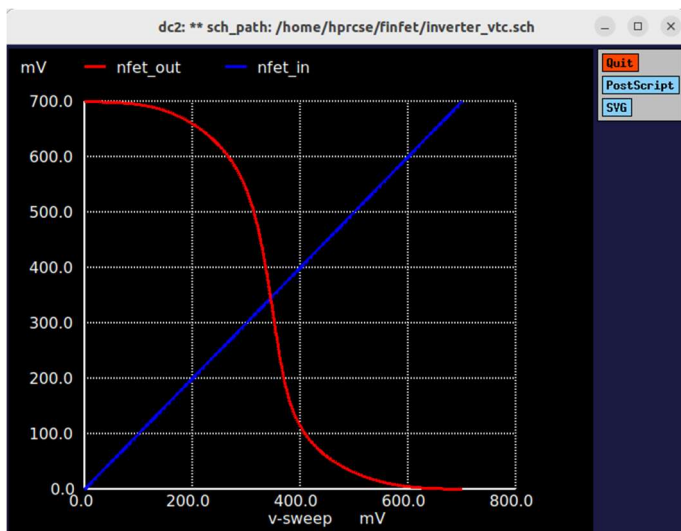
VSD 7nm FinFET Workshop

For my username, **mansi**, the ASCII sum = $109 + 97 + 110 + 115 + 105 = 536$

Therefore voltage = 0.536 mv

```
Open [ ]
1 ** sch_path: /home/hprcse/Finfet/inverter_vtc.sch
2 **.subckt inverter_vtc
3 Xpfet1 nfet_out nfet_in vdd vdd asap_7nm_pfet l=7e-009 nfin=14
4 Xnfet1 nfet_out nfet_in GND GND asap_7nm_nfet l=7e-009 nfin=14
5 V1 nfet_in GND pulse(0 0.7 20p 10p 10p 20p 500p 1)
6 V2 vdd GND 0.7
7 Vuniqu in 0 DC 0.536
8 **** begin user architecture code
9
```

- 1) Simulate the inverter using the provided SPICE deck

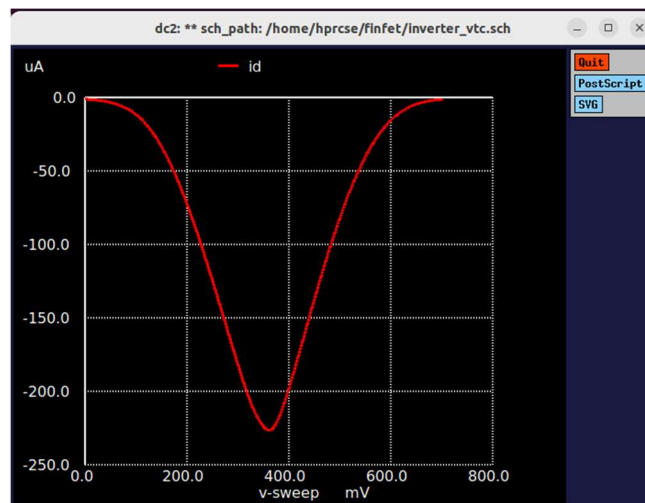


2) Modify parameters (W, L) to observe performance variation

- i) NMOS W/L = 2 ; PMOS W/L = 2
- ii) NMOS W/L = 2 ; PMOS W/L = 3
- iii) NMOS W/L = 3 ; PMOS W/L = 2

3) Extract the following metrics from simulation results

- i) $I_D = 225\mu A$



- ii) Gain, Noise Margin, Transconductance

```
Using SPARSE 1.3 as Direct Linear Solver
Reference value : 6.67000e-01
No. of Data Rows : 701
v_th      = 3.447862e-01
max_gain  = 6.428448e+00 at= 3.500000e-01
vil       = 3.488156e-01
voh       = 3.190767e-01
vil       = 3.510961e-01
voh       = 3.044187e-01
Warning from checkvalid: vector vih is not available or has zero length.
Error: RHS "voh - vih" invalid
Warning from checkvalid: vector vol is not available or has zero length.
Error: RHS "vil - vol" invalid
Warning from checkvalid: vector vih is not available or has zero length.
gm_max    = 1.235805e-03 at= 4.250000e-01
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

- iii) Power Consumption and Propagation Delay

```
Initial Transient Solution
-----
Node          Voltage
----          -
nfet_out      0.699647
nfet_in       0
vdd           0.7
in            0.536
vuniq#branch  0
v2#branch     -8.07135e-07
v1#branch     7.22294e-12

Reference value : 5.01000e-11
No. of Data Rows : 120
tpr          = 2.500000e-11
tpf          = 2.560432e-11
id_pwr       = -1.69406e-15 from= 2.00000e-11 to= 6.00000e-11
tpr = 2.500000e-11
tpf = 2.560432e-11
tp = 2.530216e-11
id_pwr = -1.69406e-15
pwr = -1.18584e-15
power = 2.964598e-05
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

- iv) Frequency

```

Using SPARSE 1.3 as Direct Linear Solver

Initial Transient Solution
-----

Node                Voltage
-----
nfet_out            0.699647
nfet_in             0
vdd                 0.7
in                  0.536
vuniq#branch        0
v2#branch           -8.07135e-07
v1#branch           7.22294e-12

Reference value : 7.50000e-11
No. of Data Rows : 71
tr = 2.100000e-11
tf = 2.351715e-11
t_delay = 4.451715e-11
f = 2.246325e+10

```

4) Characterization Table

S. No	W/L PMOS	W/L NMOS	Vth (V)	Id (A)	P (W)	tpd (ps)	Av	f (Hz)
1	2	2	3.44786e-01	225u	2.9646e-05	2.53e-11	6.428	2.246e+10
2	2	3	3.2783e-01	275u	4.0458e-05	2.52e-11	6.573	2.263e+10
3	3	2	3.6952e-01	272u	4.1032e-05	2.56e-11	6.395	2.1968e+10

Observations:

- Increasing the Pfet width shifts the VTC curve to the right.
- Increasing the Nfet width shifts the VTC curve to the left.
- Increasing the width of either the Nfet or Pfet results in a higher drain current.
- Increasing the Nfet size leads to an increase in gain.
- Increasing the Pfet size causes the gain to decrease.
- Power increases whenever the size of either device is increased.
- Drain current goes up as the drive strength increases.

